Geometry
Unit 9
Lesson 1 Practice

Name___Answer Key___
Date__________________
Period_________________

1. Rectangle $TZDP$ is 9 feet wide and 10.5 feet long. What is the area of Rectangle $TZDP$?

   $9 \times 10.5 = 94.5$ ft^2

2. Triangle RVY is 7.8 inches tall with a base of 6.2 inches. What is the area of Triangle RVY?

   $\frac{1}{2}(7.8)(6.2) = 24.18$ in^2

Find the area of each figure given the dimensions in the table.

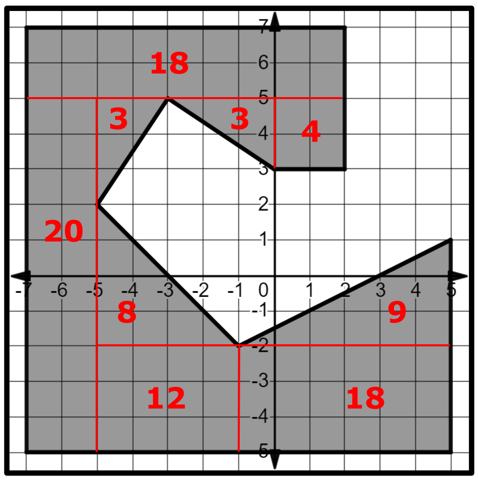
Shape	Dimensions	Area
Circle	$r = 9$	3. $\pi \cdot r^2 = \pi \cdot 9^2$ $= 81\pi$ ≈ 254.34 units^2
Parallelogram	$b = 10.2$ and $h = 4.1$	4. $(10.2)(4.1)$ $= 41.82$ units^2
Rhombus	$p = 7.3$ and $q = 7.3$	5. $\frac{(7.3)(7.3)}{2} = 26.645$ units^2
Trapezoid	$b_1 = \frac{9}{10}$, $b_2 = 8\frac{5}{6}$, $h = 9$	6. $\frac{1}{2} \cdot 9\left(\frac{9}{10} + \frac{53}{6}\right) = 43.8$ units^2

7. Kite $CMNW$ has one diagonal that is 5.4 inches and a second diagonal that is 10.8 inches. What is the area of Kite $CMNW$?

$\frac{5.4 \times 10.8}{2} = 29.16 \text{ in}^2$

Use the given diagram for questions 8-10.

Jerry is designing a boardwalk. His blueprints are shown.



8. What shapes can be used to find the area of the shaded region?

Answers may vary. Possible answers: triangles, rectangles, trapezoids, squares

9. Divide the shaded area into smaller shapes on the graph.

10. What is the area of the shaded region?

95 square units

Geometry
Unit 9
Lesson 2 Practice

Name____Answer Key__
Date_________________
Period________________

1. Pygmy Date Palm Trees are common in landscaping in Florida. Herbert bought 12 trees for his yard. Each palm tree needs approximately 16 square feet for its root system to grow. How many square feet does Herbert need of his new trees?

$12 \times 16 = 192$ ft^2

2. Herbert plans to plant a line of trees along his property line. The area he decided upon is 200 square feet. Does he have sufficient space for his 12 new trees?

Yes, 200 is more than 192.

3. If Herbert wanted to spread out the trees so that each palm tree had approximately 25 square feet, would he still have sufficient space for all 12 trees?

$25 \times 12 = 300$ sq. ft
No, 300 is more than 200.

4. What is the density of palm trees if Herbert allows only 16 square feet for each palm tree?

$\frac{1 \text{ tree}}{16 \text{ sq. ft}} = 0.0625$ tree/sq. ft

5. What is the density of palm trees if Herbert allows 25 square feet for each palm tree?

$\frac{1}{25} = 0.04$ tree/sq. ft

6. Lake George, located on the St. Johns River in Florida, covers approximately 71.88 square miles. In 2018, a survey noted there were $2{,}322$ alligators counted there. What is the density of alligators at Lake George?

$\frac{2322}{71.88} = 32.3$ alligators/mi^2

7. If 26 of the alligators counted were bull alligators, what is the density of bull alligators?

$\frac{26}{71.88} = 0.36$ bull alligators/mi^2

8. Walt Disney World spreads over 42.47 square miles. If $57{,}000$ guests enter Walt Disney World on any given day, what is the population density of visitors?

$\frac{57000}{42.47} = 1342.1$ guests/sq. mile

9. San Francisco, California covers 46.87 square miles. The population of San Francisco in 2019 was $874{,}961$. What was the population density of San Francisco in 2019?

$\frac{874961}{46.87} = 18{,}668$ people/sq. mile

10. Walt Disney World is close to the same size as San Francisco. How do the population densities compare, assuming a full day of visitors were at Walt Disney World?

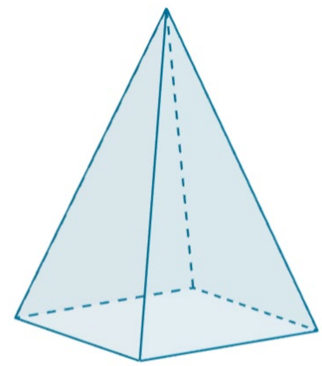
Answers will vary.
Sample answers:

- Walt Disney World is denser than San Francisco.
- $\frac{18668}{1342.1} \approx 14$; Walt Disney World is about 14 times denser in San Francisco.
- $\frac{1342.1}{18668} \approx \frac{7}{10}$; San Francisco is about $\frac{7}{10}$ as dense as Walt Disney World

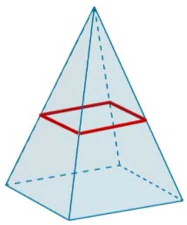
Geometry
Unit 9
Lesson 3 Practice

Name____________________
Date_____________________
Period___________________

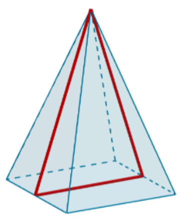
A rectangular pyramid is shown.



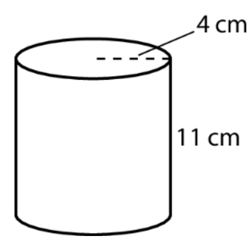
1. Draw a cross section parallel to the base.

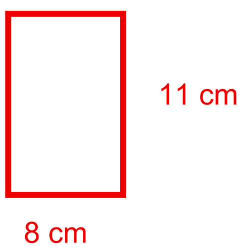


2. Draw a cross section perpendicular to the base.

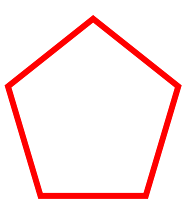


3. A cylinder is shown with height of 11 centimeters (cm) and a radius of 4 cm. Draw the cross section formed by the intersection of the cyldiner and a plane perpendicular to the base. Label the units.

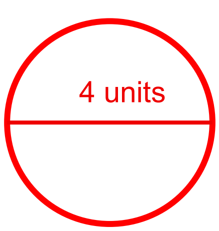




4. Draw a cross section formed by the intersection of a pentagonal prism and a plane that is parallel to the base.

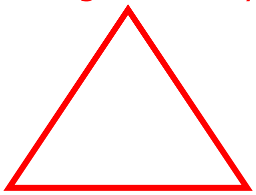


5. Draw a cross section formed by the intersection of a sphere with a diameter of 4 units and a plane passing through the center of the sphere. Label the units.



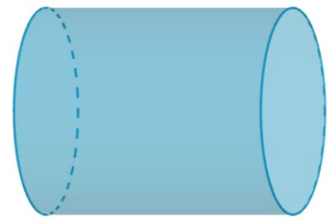
6. Draw a cross section formed by the intersection of a triangular prism and a plane parallel to its the base.

Drawings will vary.



A cylinder is shown.

7. Describe the shape of a cross section parallel to the base.

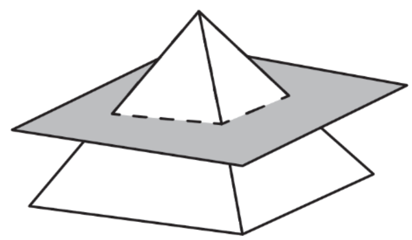


The cross section will be a circle.

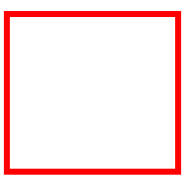
8. Describe the shape of a cross section perpendicular to the base.

The cross section will be a rectangle.

9. A square pyramid is shown. Draw the cross section formed by the intersection of the pyramid and the horiztonal plane passing through the pyramid.



Drawings will vary.



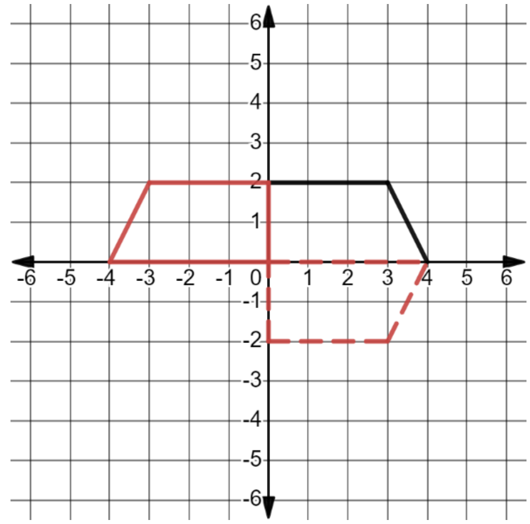
10. An ice cream cone is shown. What two-dimensional shape is created by a vertical cross section going through the ice cream cone?

The cross section will be an isosceles triangle.

Geometry
Unit 9
Lesson 4 Practice

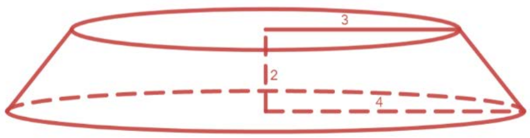
Name___Answer Key___
Date________________
Period_______________

Use the graph to answer questions 1-5.



1. Draw the transformation of the trapezoid reflected across the y-axis on the graph.
See the solid red lines on the graph

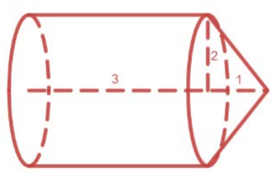
2. Draw the three-dimensional shape that would be created by rotating the trapezoid around the y-axis?



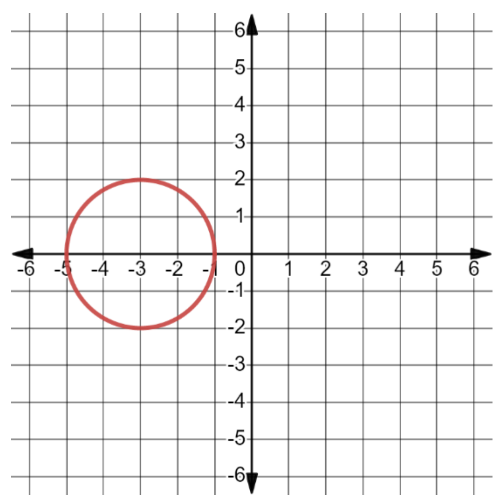
3. What are the base and height measurements of the three-dimensional shape, in units?
Length of top base = 6 units, Length of bottom base = 8 units, Height = 2 units

4. Draw the transformation of the trapezoid reflected across the x-axis in a different color on the graph.
See the dashed red lines on the graph

5. Draw the three-dimensional shape that would be created by rotating the trapezoid around the x-axis.



Use the graph of the given circle to answer questions 6-8.

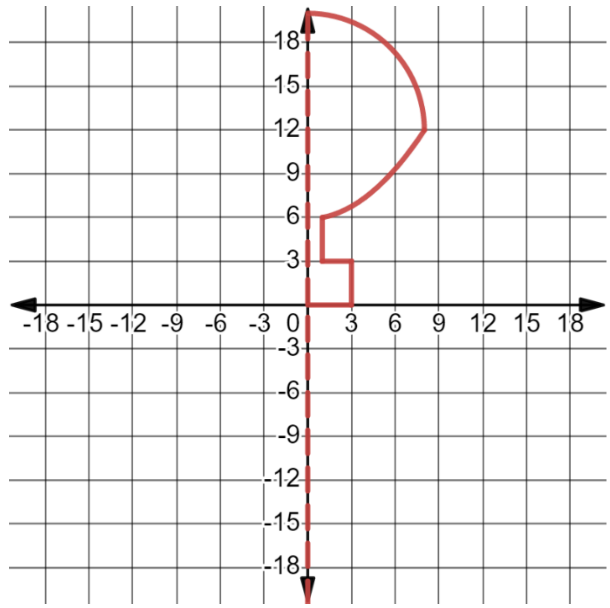
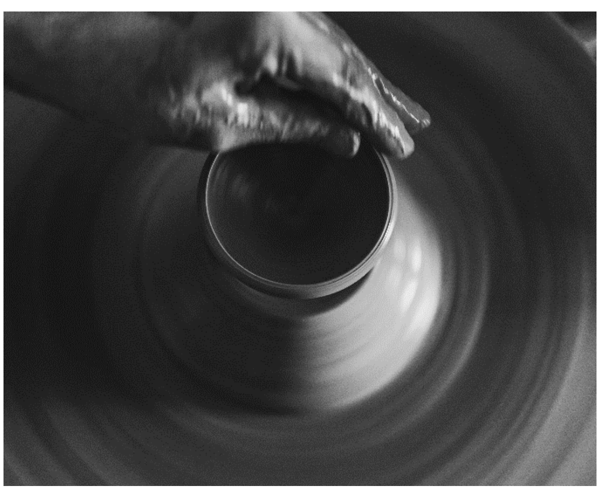
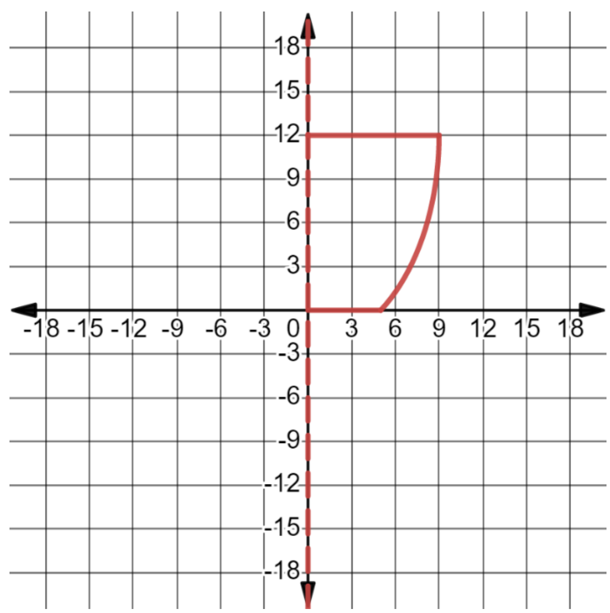


6. Draw the transformation of the circle reflected across the x-axis.
See red circle on graph (Yes, it's the same as the original black circle.)

7. What three-dimensional shape would be created by rotating the circle around the x-axis?
Sphere

8. Name one real world object that exhibits the three-dimensional shape created.
Answers will vary. Examples: basketball, soccer ball

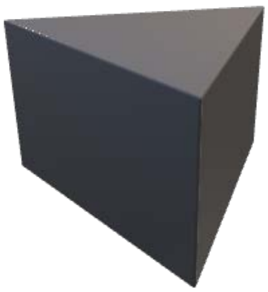
For questions 9-10, draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

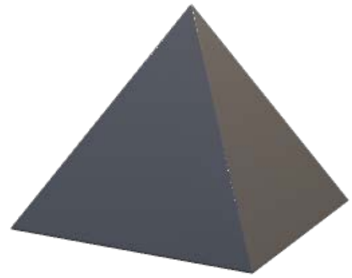
<table>
<tr><th>Image</th><th>Two-Dimensional Figure</th></tr>
<tr><td>9. A hot air balloon with a radius of 8 meters and a height of 20 meters.
</td><td>
Answers will vary. Rotated around the y-axis</td></tr>
<tr><td>10. A clay pot with a radius of 9 inches and a height of 12 inches.
</td><td>
Answers will vary. Rotated around the y-axis</td></tr>
</table>

Geometry
Unit 9
Lesson 5 Practice

Name____Answer Key__
Date_________________
Period_______________

Using the given three figures, answer questions 1-3.





1. Which two figures can have the same volume?
The cylinder and the prism.

2. Explain how the figures listed in Question 1 can have the same volume.
They can have congruent parallel cross-sections, but the pyramid cannot.

3. What information is required for each figure to determine if the volumes are equal?
Height and base dimensions

4. Miriam and Lloyd went to the movie theater. They each bought a container of popcorn. Miriam's container came with a square base. Lloyd's container had a circular base. Is it possible that they both purchased the same size of popcorn?
Yes

5. Lloyd's popcorn container was 11.5 inches high with a radius of 2.25 inches. What is the volume?
$11.5 \cdot \pi(2.25)^2 \approx 182.8$ in^3

6. Miriam's popcorn container was also 11.5 inches high but with a base whose side length was 2.25 inches. Determine if the two containers have the same volume.
$2.25(2.25)(11.5) = 58.2$ in^3
Not the same volume

7. Almond milk is sold in rectangular boxes, while evaporated milk is sold in cans. Is it possible for a given quantity of almond milk and evaporated milk to have the same volume?
Yes, if the cross-section areas are the same and the height is the same.

8. A can of evaporated milk is 4 inches tall, with a diameter of 3 inches. In a grocery store display, there are a total of 24 cans. What is the volume of milk in the display?
$4(1.5)^2(\pi)(24) = 678.24$ in^3

9. One box of almond milk is 4 inches tall, with a width of 2 inches and a length of 3 inches. There are 24 boxes in the display. What is the total volume of milk in the display?
$4 \times 2 \times 3 \times 24 = 576$ in^3

10. Compare the answers from Questions 8 and 9.
There is a larger volume of evaporated milk.

Geometry
Unit 9
Lesson 6 Practice

Name___Answer Key___
Date________________
Period_______________

1. Find the volume of a pyramid whose height is 12.9 feet and whose base is a rectangle with dimensions of 8.4 feet by 7.6 feet.

$V = \frac{lwh}{3} = \frac{(8.4)(7.6)(12.9)}{3} = \frac{823.536}{3}$

$V = 274.512\ \text{ft}^3$

2. Find the volume of a rectangular prism with a width of 8.4 feet, a length of 7.6 feet, and a height of 12.9 feet.

$V = lwh = (8.4)(7.6)(12.9) = 823.536\ \text{ft}^3$

3. What is the ratio of the answer from Question 1 to the answer from Question 2?

$\frac{274.512}{823.536} = \frac{1}{3}$ (Three pyramids make one rectangular prism.)

4. Find the volume of a sphere with a diameter of 2.82 meters. Use 3.14 for π. Round to the nearest hundredth.

$r = 1.41$

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}(3.14)(1.41)^3 = \frac{4}{3}(3.14)(2.803221) = 11.74\ \text{m}^3$

5. Find the volume of a hemisphere with a diameter of 2.82 meters. Use 3.14 for π. Round to the nearest hundredth.

$V = \frac{2}{3}\pi r^3 = \frac{2}{3}(3.14)(1.41)^3 = 5.87\ \text{m}^3$

6. What is the relationship between the answers from Questions 4 and 5?

The volume of the hemisphere is half the volume of the sphere.

7. Determine the volume of a cone with a radius of 17.1 inches, a height of 6.9 inches, and a slant height of 18.44 inches in terms of π.

$V = \pi r^2 \left(\frac{h}{3}\right) = \pi(17.1)^2 \left(\frac{6.9}{3}\right) = \pi(292.41)(2.3)$

$V = 672.543\pi$ in^3

8. Find the volume of a cylinder with a radius of 28.3 inches and a height of 14.4 inches. Round your answer to the nearest thousandth.

$V = \pi r^2 h$

$V = \pi(28.3)^2(14.4)$

$V \approx 36{,}231.410$ in^3 (when using the pi button)

9. A composite figure consists of a cylinder and a square pyramid. The height of the cylinder is 49.7 centimeters. The height of the pyramid is 26.5 centimeters. The base of the square pyramid has a width of 20 centimeters. The base of the pyramid sits on one base of the cylinder with the width of the square base equal to the diameter of the cylinder. Find the volume of the composite figure. Round to the nearest hundredth.

$d = 20$ cm

$r = 10$ cm

Volume of square pyramid: $V = \frac{lwh}{3} = \frac{(20)(20)(26.5)}{3} = 3533.33$

Volume of cylinder: $V = \pi r^2 h = \pi(10)^2(49.7) \approx 15613.72$

$3533.33 + 15613.72 = 19{,}147.05$ cm^3 (when using the pi button)

10. A composite figure consists of a cylinder and a hemisphere. The height of the cylinder is 21.4 inches. The base of the hemisphere sits on one base of the cylinder, both having a diameter of 16 inches. Find the volume of the composite figure. Use 3.14 for π. Round to the nearest hundredth.

$r = 8$

Volume of cylinder: $V = \pi r^2 h = \pi(8)^2(21.4) = 4300.544$

Volume of hemisphere: $V = \frac{2}{3}\pi r^3 = \frac{2}{3}\pi(8)^3 = 1071.787$

$V = 4300.544 + 1071.787 = 5372.33$ in^3

Geometry
Unit 9
Lesson 7 Practice

Name___Answer Key___
Date_________________
Period_________________

1. Austin lives by train tracks. He noticed a train goes by every afternoon. Most of the cars on the train are boxcars. A boxcar can be $60\frac{3}{4}$ feet long, $10\ \frac{2}{3}$ feet wide, and $11\ \frac{2}{3}$ feet tall. What is the maximum volume of a boxcar in the shape of a rectangular prism?
   $V = l \cdot w \cdot h = \left(60\frac{3}{4}\right)\left(10\frac{2}{3}\right)\left(11\frac{2}{3}\right)$
   $V = 7560\ \text{ft}^3$

2. Austin counted 21 boxcars on one train Saturday. If every boxcar was full, what was the total volume of cargo on the train?
   $21(7560) = 158{,}760\ \text{ft}^3$

3. A ping pong ball has a diameter of 40 millimeters. What is the volume of a ping pong ball? Round to the nearest hundredth.
   $r = 20$ mm
   $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(20)^3$
   $V \approx 33{,}510.32\ \text{mm}^3$ (when using the pi button)

4. The minimum diameter of a golf ball is 42.67 millimeters. What is the minimum volume of a golf ball? Round to the nearest hundredth.
   $r = 21.335$ mm
   $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(21.335)^3$
   $V \approx 40{,}678.65\ \text{mm}^3$ (when using the pi button)

5. What is the difference in volume between a golf ball and a ping pong ball?
   $40678.65 - 33510.32 = 7168.33\ \text{mm}^3$

6. Albert went on a road trip. During the drive, he noticed that some water towers were spherical, and some were cylindrical. He learned that the first water tower he saw was spherical with a 20 foot diameter. What was the volume of the first water tower? Round to the nearest hundredth.

$r = 10$

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(10)^3$

$V \approx 4188.79$ ft^3 (when using the pi button)

7. The second water tower was cylindrical with a 19 foot diameter and a 21 foot height. What was the volume of the second water tower? Round to the nearest hundredth.

$r = 9.5$

$V = \pi r^2 h = \pi(9.5)^2(21)$

$V \approx 5954.10$ ft^3 (when using the pi button)

8. Which water tower can contain more water?

The cylindrical water tower can contain more water.

9. An Olympic swimming pool is 50 meters long, 25 meters wide, and 3 meters deep. What is the volume of water in a full pool?

$V = lwh = (50)(25)(3)$

$V = 3750$ m^3

10. If the pool loses $\frac{1}{8}$ of its water depth during a free swim event, what is the new volume of water?

Step 1) $3750\left(\frac{1}{8}\right) = 468.75$

Step 2) $3750 - 468.75 = 3281.25$ m^3 left

Geometry | Name____Answer Key__
Unit 9 | Date________________
Lesson 8 Practice | Period_______________

For questions 1-10, round all answers to the nearest thousandth if necessary.

1. Find the lateral area of a pyramid whose height is 8.9 meters, slant height is 6.4 meters, and whose base is a square with a width of 7.6 meters.

$LA = \frac{1}{2}Ph_s = \frac{1}{2}(30.4)(6.4) = 97.28 \text{ m}^2$

2. Find the surface area of a pyramid with a height of 8.9 meters, slant height of 6.4 meters, and a square base with a width of 7.6 meters.

$SA = \frac{1}{2}Ph_s + s^2 = \frac{1}{2}(30.4)(6.4) + (7.6)^2 = 97.28 + 57.76 = 155.04 \text{ m}^2$

3. Find the lateral area of a rectangular prism with a width of 18.3 feet, a length of 17.5 feet, and a height of 12.4 feet.

$w = 18.3$
$l = 17.5$
$h = 12.4$
$LA = Ph = (71.6)(12.4) = 887.84 \text{ ft}^2$

4. Find the surface area of a rectangular prism with a width of 18.3 feet, a length of 17.5 feet, and a height of 12.4 feet.

$SA = 2B + Ph = 2(18.3)(17.5) + (71.6)(12.4) = 640.5 + 887.84 = 1528.34 \text{ ft}^2$

5. Find the surface area of a sphere with a diameter of 28.6 centimeters.

$r = 14.3$
$SA = 4\pi r^2 = 4\pi(14.3)^2 \approx 2569.697 \text{ cm}^2$ (when using the pi button)

6. Find the surface area of a hemisphere with a diameter of 92.5 millimeters.
$r = 46.25$
$SA = 3\pi r^2 = 3\pi(46.25)^2 = 20{,}160.189$ mm^2

7. The surface area of a sphere is 196π inches. What is the diameter of the sphere?
$SA = 4\pi r^2$
$196\pi = 4\pi r^2$
$49 = r^2$
$r = 7$ in
Diameter $= 14$ inches

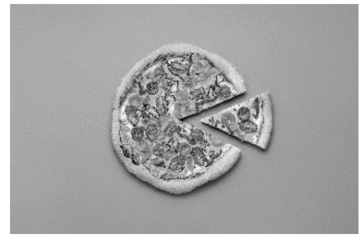
8. Find the surface area of a cone with a radius of 17.4 yards, a height of 25.3 yards, and a slant height of 30.7 yards.
$SA = \pi r^2 + \pi r h_s = \pi(17.4)^2 + \pi(17.4)(30.7)$
$= 302.76\pi + 534.18\pi \approx 2629.325$ yd^2 (when using the pi button)

9. Find the lateral area of a cylinder with a radius of 9.6 inches and a height of 8 inches.
$LA = 2\pi r h = 2\pi(9.6)(8) \approx 482.549$ in^2 (when using the pi button)

10. Find the surface area of a cylinder with a radius of 9.6 inches and a height of 8 inches.
$SA = 2\pi r^2 + 2\pi r h = 2\pi(9.6)^2 + 2\pi(9.6)(8)$
$= 184.32\pi + 153.6\pi = 1061.607$ in^2 (when using the pi button)

Geometry Name___Answer Key___
Unit 9 Date__________________
Lesson 9 Practice Period_________________

1. Rylan enjoys adding her own toppings to a frozen pizza. She bakes her frozen pizza on a pizza stone that is 12 inches in diameter and $\frac{1}{4}$ inch thick. What is the surface area of the pizza stone? Round to the nearest hundredth.

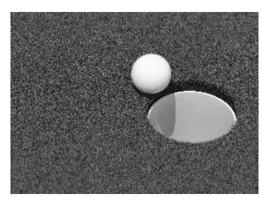


$r = 6$
$h = \frac{1}{4}$
$SA = 2\pi r^2 + 2\pi rh = 2\pi(6)^2 + 2\pi(6)(0.25) \approx 235.62$ in^2 (when using the pi button)

2. Rylan covers every possible space on top of the pizza with her favorite toppings. She only adds toppings to the top of a 12 inch diameter pizza. What is the area she has for her various toppings? Round to the nearest hundredth.
Area of circle
$A = \pi r^2 = \pi(6)^2 \approx 113.1$ in^2 (when using the pi button)

3. A ping pong ball has a diameter of 40 millimeters. What is the surface area of a ping pong ball? Round to the nearest hundredth.
Sphere
$r = 20$
$SA = 4\pi r^2 = 4\pi(20)^2 \approx 5026.55$ mm^2 (when using the pi button)

4. The minimum diameter of a golf ball is 42.67 millimeters. What is the minimum surface area of a golf ball? Round to the nearest hundredth.



$r = 21.335$
$SA = 4\pi r^2 = 4\pi(21.335)^2 \approx 5719.99$ mm^2 (when using the pi button)

5. What is the difference in surface area between a golf ball and a ping pong ball?
$5719.99 - 5026.55 = 693.44$ mm^2

6. Harry's barn needs a new coat of paint. The stable area of the barn is a rectangular prism with a base of 40 feet by 60 feet and a height of 12 feet. The loft area of the barn is a triangular prism laying on one side above the stable area. The other two sides of the triangular prism are comprised of roof shingles, so the only portion that needs paint are the two base triangles. The height of the loft is 10 feet, with a base width of 40 feet. Find the area of the barn that needs to be painted.

Lateral area of rectangular prism plus two triangles

$(200)(12)+\frac{1}{2}(10)(40)(2)=2400+400=2800$ ft^2

7. If 1 gallon of paint covers 400 square feet, how much paint does Harry need to buy?

$\frac{2800}{400}=7$ gallons of paint

8. If 1 gallon of paint costs $\$20.90$, how much will the paint cost?

$7(20.90)=\$146.30$

9. Bert wants to build a new pen for his chickens. He has one area of his property available for a rectangular pen that would measure 20 feet by 35 feet. He has a second area of his property available for a square pen that would measure 27 feet wide. Which pen area would require the least amount of fence?

Rectangular fence: $2(20+35)=110$ ft

Square fence: $(27)(4)=108$ ft

The square fence would require the least amount of fence.

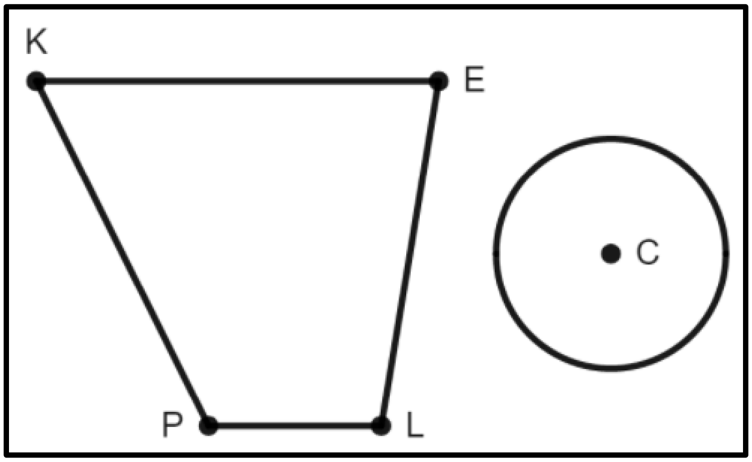
10. Bert decides to build the pen area with the least amount of fence. If his fence is 4.5 feet tall, what is the surface area of his fence?

$Ph=(108)(4.5)=486$ ft^2

Geometry
Unit 9
Lesson 10 Practice

Name___Answer Key___
Date________________
Period_______________

A diagram with Trapezoid $KELP$ and Circle C is shown. Given the dimensions, find the area of each figure and complete the table. Round to the nearest hundredth if necessary.



Shape	Dimensions	Area
Trapezoid $KELP$	$h = 14.2,\ b_1 = 13.7,\ b_2 = 5.6$	1. $\frac{1}{2}(13.7 + 5.6)(14.2)$ $= 137.03$ units^2
Circle C	$r = 9.8$	2. $\pi(9.8)^2$ ≈ 301.72 units^2 (when using the pi button)

Determine the area of the dilated figures and complete the table.

Shape	Area with Scale Factor of $\frac{5}{2}$	Area with Scale Factor of $\frac{2}{5}$
Trapezoid $K'E'L'P'$	3. $(137.03)\left(\frac{25}{4}\right) = 856.44$ units^2	4. $(137.03)\left(\frac{4}{25}\right) \approx 21.92$ units^2
Circle C'	5. $(301.72)\left(\frac{25}{4}\right) = 1888.75$ units^2	6. $(301.72)\left(\frac{4}{25}\right) \approx 48.28$ units^2

7. Parallelogram $KEYS$ has a base of 8.5 feet and a height of 12.5 feet. What is the area of Parallelogram $KEYS$?

$A = bh = (8.5)(12.5) = 106.25 \text{ ft}^2$

8. Parallelogram $KEYS$ is multiplied by a scale factor to create Parallelogram $K'E'Y'S'$. Parallelogram $K'E'Y'S'$ has an area of 425 square feet. What is the scale factor for the area of the pre-image to get the image?

$\frac{425}{106.25} = 4$ area ratio

Scale factor of dimensions $= 2$

9. Using the scale factor, what is the base of the image?

$8.5 \times 2 = 17$ ft

10. Using the scale factor, what is the height of the image?

$12.5 \times 2 = 25$ ft

Geometry
Unit 9
Lesson 11 Practice

Name___Answer Key___
Date__________________
Period__________________

1. When Cheryl was young, she had a twin-size bed. It was 38 inches by 75 inches. What was the base (sleeping) area of the twin bed?
   $A = (38)(75) = 2850$ in^2

2. As Cheryl grew taller, she needed a bigger bed. Cheryl purchased a twin XL bed. It was 38 inches by 80 inches. What was the base (sleeping) area of the twin XL bed?
   $A = (38)(80) = 3040$ in^2

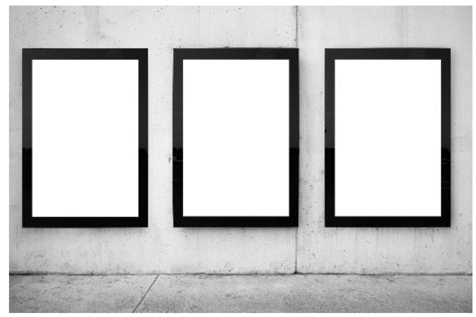
3. What is the difference in areas of the two beds?
   $3040 - 2850 = 190$ in^2

4. With the bed frame, both of Cheryl's beds stood 25 inches high. What was the lateral surface area of the twin bed?
   $\big(2(38 + 75)\big)(25) = (226)(25) = 5650$ in^2

5. What was the lateral surface area of the twin XL bed?
   $\big(2(38 + 80)\big)(25) = (236)(25) = 5900$ in^2

6. Cheryl's grandma made a quilt for the twin XL bed. It was big enough to cover the bed and touch the floor on all four sides. What was the area of the quilt?
$3040 + 5900 = 8940$ in^2

7. There are three posters on display at the corner store. One is 18 inches by 24 inches, the second is 24 inches by 36 inches, and the third is 27 inches by 40 inches. What is the area of the smallest poster?
$18 \times 24 = 432$ in^2



8. What is the area of the middle size poster?
$24 \times 36 = 864$ in^2

9. What is the area of the largest poster?
$27 \times 40 = 1080$ in^2

10. David designated space on his wall for posters that is 50 inches by 75 inches. Can he fit all 3 posters in his designated space?
It will fit by area, because $50 \times 75 = 3750$ in^2, but he will have to cut the posters to make the length and width fit, so no.